228 (2023) 211924

Contents lists available at ScienceDirect

Geoenergy Science and Engineering

journal homepage: www.sciencedirect.com/journal/geoenergy-science-and-engineering

Risk prediction for petroleum exploration based on Bayesian network classifier

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ARTICLE INFO

Keywords:
Oil and gas energy
Petroleum exploration
Bayesian network classifier
Risk prediction
Decision-making

ABSTRACT

Effective prediction of petroleum exploration risks is critical for boosting oil and gas extraction efficiency and increasing economic benefits. To improve the accuracy and efficiency of risk prediction, this paper proposes a scalable k-dependence Bayesian network classifier (SKDB) method for solving the risk prediction problem of oil and gas exploration. The effectiveness and efficiency of the SKDB method were illustrated by taking the Jurassic Sangonghe Formation hydrocarbon exploration risk prediction in the Junggar Basin as an example. The model accuracy comparison results showed that SKDB has the highest accuracy compared with the state-of-the-art methods, and the accuracy was improved by 3.55~7.63%. The oil and gas probability map prediction results indicated that the SKDB method not only showed high hydrocarbon probability in the reservoir areas, but also had better trend inference ability compared with other methods. Based on the SKDB prediction results, the remaining hydrocarbon favorable areas of the Sangonghe Formation reservoir were visualized and three types of favorable hydrocarbon distribution zones were preferentially selected, namely Type I (highly favorable), Type II (moderately favorable) and Type III (low favorable). The application results can provide a decision basis for the next oil and gas exploration and optimization strategy.

1. Introduction

Every year, oil companies invest billions of dollars in petroleum exploration. Since 1980, China National Petroleum Corporation (PetroChina) has drilled thousands of new exploratory wells each year, with an annual average drilling success rate of only 42.6% (An exploratory well that produces industrial oil or gas flow in the target layer is a productive well, otherwise is a non-productive well) (Guo et al., 2019). This means that there is still a great deal of room for improvement in terms of the exploration success rate. Without considering the economic investment, the presence or absence of oil and gas in a particular space is a decision-making problem related to risk and uncertainty. Geological risk prediction, i.e., accurate spatial prediction of hydrocarbon distribution, is an essential part of petroleum exploration and has vital significance for improving exploration efficiency, optimizing drilling strategies, and enhancing economic benefits (Xie et al., 2011; Roisenberg et al., 2009; Zhu et al., 2018).

Geological risk prediction usually examines all essential geological factors (e.g., source rock, reservoir, cap rock, etc.), which are significant for oil or gas accumulation. Experts comprehensively superimpose various geological factors on a risk map, which reflects the probability of hydrocarbon occurrence, by giving different probabilities for different geological factors (MacKay, 1996; Otis and Schneidermann, 1997; White, 1988, 1993; Rose, 2001). However, conventional geological risk evaluation methods have some drawbacks. Firstly, the repeatability and consistency of subjective judgments used in risk assessment are weak, because the probability values given by different experts to different geological factors may be very considerable, while a consistent and repeatable risk assessment result is more desirable for consistent petroleum exploration analysis (Baddeley et al., 2004). Secondly, the traditional evaluation method fails to integrate the spatial correlation of different prospects in the prediction of geological risks. Since the occurrence of hydrocarbons in a specific exploration area is not an independent event, but the result of the combination of various geological factors that constitute the hydrocarbon system, it is highly critical to correctly understand the spatial variation of exploration objects and the spatial correlation with geological factors for geological risk

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https://doi.org/10.1016/j.geoen.2023.211924
Received 1 October 2022; Received in revised form 3 April 2023; Accepted 13 May 2023
Available online 7 June 2023

prediction (Hu et al., 2009; Chen et al., 2008). Geological risk can be considered as spatial random variables, and obtaining its trends can help describe the heterogeneity of hydrocarbon occurrence and enhance exploration efficiency (Zhu et al., 2018).

In the past two decades, many experts and scholars have considered the spatial correlation of exploration targets and have made extensive attempts to systematically predict the risk of oil and gas occurrence at a given prospect (Rose, 2001; Suslick et al., 2009; Roisenberg et al., 2009; Kim and Lee, 2018). For example, Gao et al. (2000) introduced an object-based computational model for the spatial distribution of oil and gas at a play level. Chen and Osadetz (2006b), Chen et al. (2000, 2004a,b, 2008), and Hu and Guo (2007) described a model-based method to characterize the spatial correlation of geological factors and evaluate the probability of undiscovered hydrocarbon resources. Tounsi (2005) and Roisenberg et al. (2009) used a fuzzy method to analyze the risk of oil exploration. Xie et al. (2011), Chen and Osadetz (2006a), Handhal et al. (2019), and Zhu et al. (2018) adopted a statistical method to form a hydrocarbon probability map, which can reflect the synthesis of various geological data, geophysical data, exploration, and drilling results, etc. Although the above methods have achieved certain results, most of these methods are modeled from the perspective of discrimination and cannot quantitatively express the potential relationship between different geological attributes. As the most popular probabilistic graphical model, Bayesian network classifier (BNC) can not only graphically describe the qualitative relationship between attributes but also quantitatively describe the dependency relationship between attributes. In recent years, BNC has been used to solve oil and gas exploration problems. For example, Ren et al. (2020) proposed to use of the tree augmented Bayesian network to integrate geological factors to characterize the spatial distribution of petroleum. Guo et al. (2022) introduced a Bayesian ensemble model to predict favorable accumulation areas of hydrocarbon resources.

Although the current state-of-the-art BNC prediction method has achieved good results, there are still two shortcomings: (i) The dependency relationships between different attributes are not sufficiently portrayed in the model construction process. (ii) The accuracy of the model and the prediction effect have more room for improvement. To improve the shortcomings of the above methods, this paper proposes a novel method to predict the risk of oil and gas exploration. The main novelties of this paper are as follows: (i) Understanding the oil and gas geological conditions of the Sangonghe Formation from multiple aspects such as oil and gas sources, reservoir burial depth, sedimentary environment, trap conditions, and preservation conditions. (ii) A scalable k-dependent Bayesian network classifier (SKDB) method is applied for the first time to solve the oil and gas exploration risk prediction problem. (iii) The SKDB algorithm is superior to other state-of-the-art methods in terms of both accuracy and application effect. (iv) The prediction results can provide a quantitative decision basis for the selection of future hydrocarbon exploration targets and the deployment of exploration wells.

The remainder of this study is structured as follows: Section 2 introduces the method principle and Bayesian network model. In Section 3, a case study with the proposed method is illustrated. Section 4 draws the conclusions.

2. Methods

2.1. Principles of risk prediction for petroleum exploration

Petroleum exploration is an engineering problem with uncertainty. A suitable geological model to describe the occurrence of reservoir accumulations can effectively reduce exploration risks. Various geologists and researchers have applied high levels of expertise to answer issues such as: if the prospect has a chance to find a field? Where and how much petroleum resources are accumulated? Which method is used to assess petroleum resources? Capturing geological knowledge and developing an appropriate model to solve the aforementioned problems is a permanent goal in petroleum companies (Tounsi, 2005; Roisenberg et al., 2009). In the present study, the problem of predicting geological risk is investigated. The purpose of exploration geological risk prediction is to identify whether potential drilling in a target layer will be productive or non-productive according to our knowledge of the basin and existing data. That is, we expect to learn the chance of successfully finding hydrocarbon in a given reservoir before drilling. Therefore, the oil and gas exploration risk analysis can be transformed into the problem of binary classification based on multiple characteristics in machine learning in this paper (Chen et al., 2008).

Assuming that the nature of the target layer in the hydrocarbon-bearing areas can be divided into two groups, namely, producer area and non-producer area. n exploration wells have been drilled in the petroleum play. Actual exploration has shown that the producer and the non-producer areas in the same reservoir have different geological conditions. Therefore, the premise of classification is the assumption that the characteristics of different groups are statistically significantly different, while there are no obvious differences within the same group. Based on exploratory well results and other oilfield data, the probability that an undrilled location falls into one of the two defined groups is estimated to visualize geological risks and optimize drilling strategy. Meanwhile, this estimation can be expressed in the conditional probability form. The exploratory well result can be represented by a random binary variable C, with $c = 1$ and $c = 0$ representing the exploratory wells located in the producer and non-producer zones, respectively. It is assumed that whether each exploratory well is a productive well is associated with m geological variables, and the vector $\mathbf{X} = (X_1, X_2, \ldots, X_m)$ to represent m geological feature information that decides whether a potential drilling is a productive well or a non-productive well. Thus, the exploratory well category variable C has a dependence on each geological attribute variable $X_k(k = 1 \ldots m)$, but the different geological attribute variables may be dependent or independent of each other, which needs to be analyzed according to the actual exploration results in different study areas since different basins have distinct tectonic evolution histories, which leads to different roles of the main factors controlling hydrocarbon generation. The conditional probability that the area at location $\mathbb{L}$ belongs to the productive area for given observations $\mathbf{x} = (x_1, x_2, \ldots, x_m)$ is written as follows

$$P(c = 1|\mathbf{x}) = \frac{P(c = 1, \mathbf{x})}{P(\mathbf{x})} = \frac{P(x_1, x_2, \ldots, x_m, c = 1)}{P(x_1, x_2, \ldots, x_m)} \tag{1}$$

According to the chain decomposition of the joint probability distribution and the Bayesian formula, Eq. (1) is transformed as follows

$$\begin{aligned} P(c = 1|\mathbf{x}) &= \frac{P(c = 1)P(\mathbf{x}|c = 1)}{P(\mathbf{x})} \\ &= \frac{P(c = 1)P(x_1|c = 1)P(x_2|c = 1,x_1), \ldots, P(x_m|c = 1,x_1,x_2, \ldots, x_{m-1})}{P(x_1)P(x_2|x_1), \ldots, P(x_m|x_1, \ldots, x_{m-1})} \end{aligned} \tag{2}$$

The spatiotemporal complexity of computing Eq. (2) grows exponentially as the variable m increases. To avoid the complexity of calculation, this study uses a novel BNC method to integrate all available geological factors and assess oil and gas exploration risk. The next section will introduce the knowledge of the Bayesian network.

2.2. Bayesian network and Bayesian network classifier

Bayesian network (BN) is a two tuple $\mathcal{B} = (G, \Theta)$, where $G = (\mathbf{V}, \mathbf{E})$ is a directed acyclic graph (DAG), also known as a BN structure, $\mathbf{V} = \{X_1, X_2 \cdots X_n\}$ is a node set, and $\mathbf{E} = \{(X_i \rightarrow X_j) \mid X_i, X_j \in \mathbf{V}\}$ is a directed edge set, $\Theta = \{\theta^1, \ldots, \theta^n\}$ is the conditional probability distribution table corresponding to the random variable $\mathbf{V}$ in G, and the conditional probability distributions for each variable have the following form

$$\theta^i = P\left(X_i \mid Pa\left(X_i\right)\right) \tag{3}$$

where $Pa(.)$ denotes the set of parent nodes of the corresponding variables. In the BN structure, if a directed edge $(X_i \rightarrow X_j)$ exists between any two nodes X_i and X_j, then X_j is the child of X_i and X_i is the parent of X_j.

BNC is a special type of BN designed for classification problems (Bielza and Larranaga, 2014; Ren and Guo, 2023), where in addition to the attribute nodes, the network also has a special class node. It is general to use $\mathbf{V}' = \left\{X_1, X_2 \cdots X_n, C\right\}$ to represent all nodes in the BNC structure, where $\mathbf{X} = \left\{X_1, X_2 \cdots X_n\right\}$ is the set of attribute features used to predict class labels and C is the class label (node). According to the Markov property of BN, the joint probability distribution $P(\mathbf{V}')$ is calculated as follows

$$P(\mathbf{V}') = P\left(X_1, \ldots, X_N, C\right) = P(C|Pa(C)) \prod_{i=1}^{N} P\left(X_i \mid Pa\left(X_i\right)\right) \tag{4}$$

Assuming that there is BNC $\mathcal{B}$, for a sample(instance) to be classified with n attributes, $\mathbf{x} = \left\{x_1, x_2, \ldots, x_n\right\}$, the classification principle of BNC is to calculate all possible class label probabilities for the tested sample through Bayes' theorem, and assign the class value that has the highest posterior probability to the tested sample $\mathbf{x}$. The calculation equation of the predicted class label c^* is as follows

$$c^* = \arg\max_{c \in \Omega_C} P(c \mid \mathbf{x}) = \arg\max_{c \in \Omega_C} \frac{P(\mathbf{x}, c)}{P(\mathbf{x})} \tag{5}$$

As $P(\mathbf{x}) = \sum_{c \in \Omega_C} P(c, \mathbf{x})$, computing the posterior probability $P(c \mid \mathbf{x})$ is the same as computing $P(\mathbf{x}, c)$. Combining Eqs. (4) and (5), Assuming that $Pa(C) = \emptyset$, the following classification rule can be obtained

$$\begin{aligned} c^* &= \arg\max_{c \in \Omega_C} P(c \mid \mathbf{x}) \\ &\propto \arg\max_{c \in \Omega_C} P(c, \mathbf{x}) \\ &\propto \arg\max_{c \in \Omega_C} P(c) \prod_{i=1}^{n} P\left(x_i \mid Pa\left(x_i\right)\right) \end{aligned} \tag{6}$$

Through the above analysis, Eq. (2) is derived as follows

$$P(c=1|\mathbf{x}) \propto P(c=1) \prod_{i=1}^{n} P\left(x_i \mid Pa\left(x_i\right)\right) \tag{7}$$

Therefore, we only need to determine the parent node set $Pa(X_i)$ of each attribute variable X_i, that is, Eqs. (6) and (7) can be calculated by determining BNC structure. Next subsection, we will introduce a scalable K-dependence Bayesian network classifier.

2.3. The SKDB algorithm

The SKDB algorithm mainly consists of two parts: attribute sorting and model simplification (Ren and Wang, 2020).

2.3.1. Attribute sorting

The purpose of attribute sorting is to reduce the computational complexity, that is, if node X_i is before node X_j, X_j cannot become a parent node of X_i. Specifically, the set of possible parent nodes of node X_i can only contain those nodes that are ahead of it in the input sequence. The attribute sorting process of SKDB mainly uses mutual information chain rules to sort attributes.

From an information theory point of view, in the process of attribute sorting, the set composed of the attributes to be selected and the selected attributes and class variables should have the largest mutual information (MI) value, that is, the strongest dependency. Specifically, suppose $\boldsymbol{S}_{i-1} = \left\{X_1, X_2, \ldots, X_{i-1}\right\}$ represents a set of the $i-1$ attribute nodes that have been chosen during attribute sorting. Our purpose is to maximize the MI value $MI(C; X_i, \boldsymbol{S}_{i-1})$ when selecting the ith attribute. According to the chain rule of MI, the process of maximizing $MI\left(C; X_i, \boldsymbol{S}_{i-1}\right)$ is derived as shown below

$$\begin{aligned} \max MI(C; X_i, \boldsymbol{S}_{i-1}) &= MI(C; \boldsymbol{S}_{i-1}) + CMI(C; X_i|\boldsymbol{S}_{i-1}) \\ &\propto \max CMI(C; X_i|\boldsymbol{S}_{i-1}) \\ &= MI(C; X_i) - MI(X_i; \boldsymbol{S}_{i-1}) + CMI(X_i; \boldsymbol{S}_{i-1}|C) \\ &= MI(C; X_i) \\ &\quad - \sum_{X_j \in \boldsymbol{S}_{i-1}} (MI(X_i; X_j) - CMI(X_i; X_j|C)) \end{aligned} \tag{8}$$

where the MI and conditional mutual information (CMI) are defined as follows

$$MI(X_i; C) = \sum_{x_i, c} P(x_i, c) \log \frac{P(x_i, c)}{P(x_i)P(c)} \tag{9}$$

$$CMI(X_i; X_j|C) = \sum_{x_i, x_j, c} P(x_i, x_j, c) \log \frac{P(x_i, x_j|c)}{P(x_i|c)P(x_j|c)} \tag{10}$$

It can be seen from Eq. (8) that when using this equation to sort attributes, not only the $MI(C; X_i)$ is used to consider the relationship between class node and geological attribute nodes, but also the $MI(X_i; X_j)$ and $CMI(X_i; X_j|C)$ are used to consider the (conditional) dependency between geological attribute nodes. Therefore, the attribute sequence sorted according to this criterion can maximize the dependency between class variable and geological attribute variables, thus effectively improving the classification performance.

The algorithm 1 gives the pseudo-code of attribute sorting in SKDB model.

Algorithm 1 Attribute sorting.

Input: Dataset $\mathcal{D}$.
Output: Sorted sequence $\boldsymbol{S} = \{X_1, \ldots, X_n\}$.
1: Calculate $CMI(X_i; X_j|C)$, $MI(X_i; X_j)$, and $MI(X_i; C)$.
2: Unsorted attribute sequence $\boldsymbol{X} = \{X_n, \ldots, X_1\}$, $\boldsymbol{S} = \{\}$.
3: **for** $i = 1$ to n **do**
4: Choose the attribute X_i that maximizes $MI(C; X_i, \boldsymbol{S}_{i-1})$, and $X_i \in \boldsymbol{X}$.
5: Remove X_i from $\boldsymbol{X}$.
6: Add X_i to $\boldsymbol{S}$.
7: **end for**
8: **return** $\boldsymbol{S}$.

2.3.2. Model simplification

In the process of model simplification, the initial network structure is established by using CMI, and then an iterative search dynamic filtering mechanism is used to simplify the model and delete redundant edges. The threshold T is calculated as follows

$$T = \mu - \sigma * it/MaxIt \tag{11}$$

where, μ and σ denote the average and standard deviation of the CMI between all connected attribute nodes in the network structure, respectively, it represents the current number of iterations, and $MaxIt$ represents the maximum number of iterations. When the weight of the edge between two attributes is less than a certain threshold, the two attributes are considered to be less dependent, and the edge is deleted. Formally, the $CMI\left(X_i; X_j \mid C\right)$ is taken to evaluate the weight between any two connected attributes X_i and X_j. If $CMI\left(X_i; X_j \mid C\right) < T$, remove the edge $(X_i \rightarrow X_j)$ from the network. After traversing the edges between all attributes and filtering, the model is output and its classification performance is verified. When the current iteration number of iterations it reaches $MaxIt$, the optimal structure is output. Algorithm 2 is the pseudo-code of the filtering mechanism in SKDB.

Algorithm 2 Filtering.

Input: Dataset D, the sorted attribute sequence $\boldsymbol{S}$, and K value.
Output: The topological structure of SKDB, G.
Initialization: $G = (\mathbf{V}, \mathbf{E})$ where $\mathbf{V} = \{C\}$, $\mathbf{E} = \{\}$, and $\boldsymbol{S} = \{X_1, \ldots, X_n\}$.
for $i = 1$ to n **do**
  for $j = 1$ to $(i-1)$ **do**
    $\mathbf{E} = \mathbf{E} \cup (C \rightarrow X_i)$, $\mathbf{V} = \mathbf{V} \cup X_i$.
    Choose $d = min(i-1, K)$ parent nodes for X_i, and each parent node X_j satisfies so that the value of $CMI(X_i; X_j|C)$ is maximized.
    $\mathbf{E} = \mathbf{E} \cup (X_j \rightarrow X_i)$.
  end for
end for
for it to $MaxIt$ **do**
  $G_{temp} = G$, and calculate μ and σ of the weights on all edges in G.
  for each edge $(X_j \rightarrow X_i)$ in G_{temp} **do**
    if $CMI(X_i; X_j|C) < T$ **then**
      Remove edge $(X_j \rightarrow X_i)$ from G_{temp}, that is, $\mathbf{E} = \mathbf{E} \backslash (X_j \rightarrow X_i)$.
    end if
  end for
  $\boldsymbol{LVD}(it) = LossVD(G_{temp})$ // $LossVD(G_{temp})$ represents the loss of the evaluation network G_{temp} on the validation set.
end for
  Choose the topological structure corresponding to the lowest value in $\boldsymbol{LVD}$, denoted as G_{best}.
return $G = G_{best}$.

2.3.3. The SKDB algorithm

According to the above two sections, algorithm 3 gives the pseudo-code of the SKDB algorithm used in this paper, and the main procedures of the SKDB algorithm are shown below:

Step 1: Initialize the parameters, including dataset D, K, and $MaxIt$.

Step 2: Sort attributes according to algorithm 1, as shown in Fig. 1(a).

Step 3: Simplify the network model according to algorithm 2, as shown in Figs. 1(b)–(c).

Step 4: Find the optimal network structure (Fig. 1(d)) and learn the parameters.

Step 5: Output the classifier.

Algorithm 3 SKDB.

Input: Dataset D, K value, and $MaxIt$.
Output: BNC: SKDB.
$\boldsymbol{S} = Sorting$ // Algorithm 1
$G = Filtering$ // Algorithm 2
Let Θ denotes the set of parameters and calculate Θ based on G and D.
$SKDB = (G, \Theta)$
return $SKDB$.

3. Application example

In this section, the Jurassic Sangonghe Formation in the hinterland of the Junggar basin is taken as an application example, and the SKDB model is used to predict the hydrocarbon variation law of the Sangonghe Formation, so as to illustrate the effectiveness of the SKDB method in solving practical application problems. Meanwhile, the exploration risk is visualized according to the prediction results, and the favorable distribution areas of remaining oil and gas resources in the Sangonghe Formation are pointed out, so as to provide a decision-making basis for hydrocarbon exploration and development.

3.1. Geological setting

The application is selected from the central part of the Junggar Basin and covers an area of about 2.7×10^4 km^2, including the eastern part of the Mahu Sag, Dongdaohaizi Sag, Dabasong Uplift, Mosuowan Uplift, Mobei Uplift, and Luxi Slope, etc. The hydrocarbon-bearing rock is the Permian Lower Wuerhe Formation, the reservoir is the Jurassic Sangonghe Formation, i.e., the target layer, and the cap rock is the Jurassic mud shale. By the end of 2019, 203 exploratory wells in the target layer have been drilled, and the discovered petroleum and natural gas reserves are 1.128×10^8 t and 459×10^8 m^3 respectively. Fig. 2 depicts the geographical location of the research area and the distribution of major oil fields.

3.2. Key parameters extraction

Based on the information of exploratory wells penetrating the Sangonghe Formation, and the petrophysical data, reserves, sedimentary facies maps, seismic structure maps, fault maps, trap maps, and basin modeling results of 54 reservoirs, five geological factors were extracted for predicting the exploration risk in the study area.

(1) Hydrocarbon supply conditions

These conditions indicate the probability of hydrocarbon supply, which is defined to be from 0 (probability of 0) to 1 (probability of 100%). The probability distribution is determined in terms of the hydrocarbon-generating intensity map, the map of faults connecting underlying hydrocarbon-bearing rocks, and the modeled hydrocarbon migration pathways (Guo et al., 2018). Fig. 3(a) is the quantitative evaluation map of hydrocarbon supply.

(2) Reservoir conditions

These conditions indicate the probability of reservoir rocks occurrence, which was established based on the Sangonghe sedimentary facies (fluvial-delta facies) and petrophysical data from the reservoirs. Fig. 3(b) is the quantitative evaluation map of Sangonghe reservoir.

(3) Trap conditions

These conditions indicate the probability of trap occurrence, which was set to be 100% for the areas with discovered oil and gas reservoirs, 70%–80% for confirmed traps, 60%–70% for the traps to be confirmed, 30%–60% for prognostic lithologic traps, and < 30% for additional areas. Fig. 3(c) is the quantitative evaluation map of trap conditions.

(4) Caprock and preservation conditions

These conditions indicate the probability of hydrocarbon sealing and preservation, which was determined depending on overlying faults and unconformable weathered zones. The probability is set to be 10%–20% in the area where overlying faults are unfavorable for hydrocarbon preservation, 70%–90% in the area where unconformable weathered clays function as good seals for hydrocarbons, and 40%–60% in the area where mud shales serve as local seals. Fig. 3(d) is the quantitative evaluation map of caprock and preservation conditions.

(5) Burial depth of reservoir top interface

The buried depth of the top interface of the Sangonghe Formation is determined through two-dimensional seismic reflection profiles, drilling, and literature data. Meanwhile, the buried depth data in the study area were normalized using the following equation

$$X = \frac{X' - X_{min}}{X_{max} - X_{min}} \tag{12}$$

where X is the normalized depth data, X' represents the initial depth data, X_{min} represents the minimum value of the original depth data and X_{max} represents the maximum value of the original depth data. Fig. 3(e) is a normalized distribution map of burial depth.

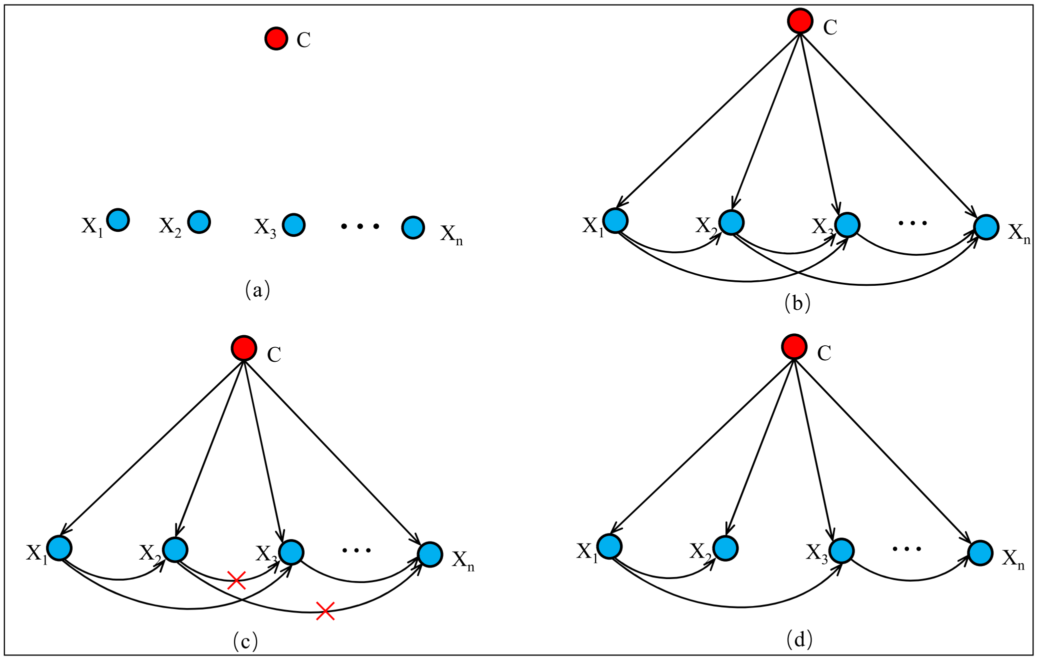

Fig. 1. Schematic diagram of SKDB topology construction process((a) ordered attribute, (b) initial SKDB structure, (c) delete redundant edges, (d) best structure).

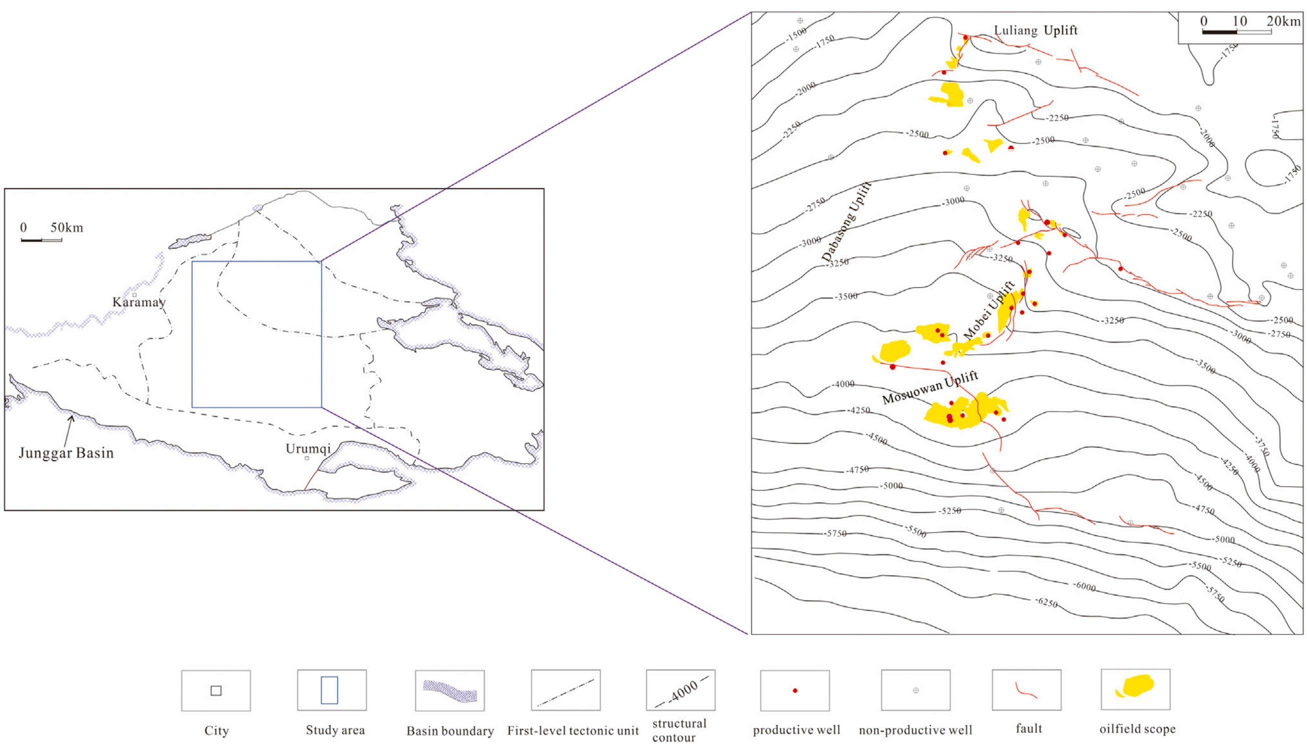

Fig. 2. Location of the study area.

3.3. *Dataset construction*

(1) Exploratory well dataset

We first collected data on the results of 203 exploratory wells drilled through the Sangonghe Formation, including 94 non-productive and 109 productive wells, respectively. Then, the five key geological parameters in Fig. 3 were collected sequentially on each well, resulting in a class-labeled dataset with 203 samples (Table 1).

(2) Grid dataset

In addition to the exploratory well dataset, we used the two-dimensional Perpendicular Bisection grid technology to construct about 13,000 evaluation units, and also collected five kinds of key geological parameters in each evaluation unit to form a grid dataset(Table 2).

3.4. *Comparison method and evaluation criteria*

(1) Comparison method

To illustrate the effectiveness and superiority of the SKDB method, we compared it with the following five state-of-the-art methods for predicting oil and gas exploration risks:

• TAN: Tree augmented Bayesian network method (Ren et al., 2020).

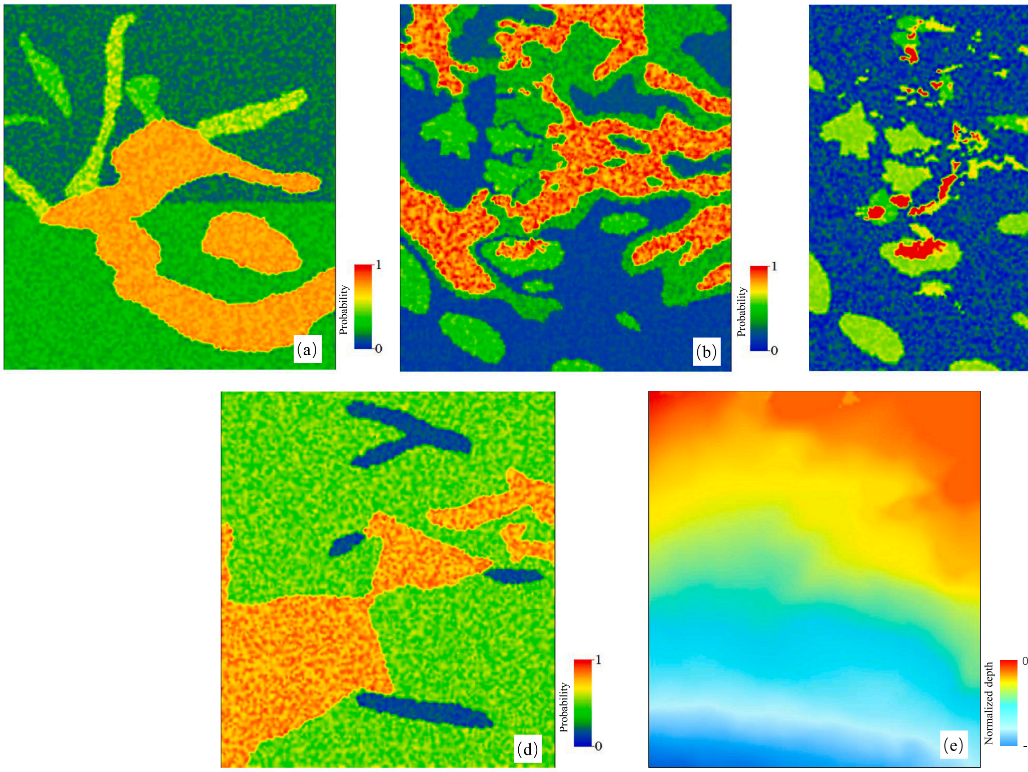

Fig. 3. Key geological parameters ((a): Hydrocarbon source conditions; (b): Reservoir conditions; (c): Trap conditions; (d) Caprock and preservation conditions; (e) Normalization of burial depth).

Table 1
Dataset of 203 exploratory wells.

Number	HSC	RC	TC	CAPC	ND	Well
1	0.1	0.45	0.13	0.4	−0.25	0
2	0.18	0.37	0.07	0.51	−0.24	0
3	0.4	0.12	0.14	0.5	−0.28	0
4	0.47	0.53	0.54	0.58	−0.32	0
5	0.44	0.41	0.14	0.5	−0.21	1
6	0.16	0.87	0.17	0.57	−0.17	0
7	0.3	0.52	0.56	0.42	−0.33	0
8	0.16	0.17	0	0.54	−0.25	0
9	0.14	0.77	0.02	0.41	−0.21	0
10	0.72	0.9	0.06	0.18	−0.19	0
11	0.14	0.44	0.17	0.7	−0.18	0
12	0.15	0.22	0.15	0.81	−0.17	0
13	0.74	0.79	0.19	0.14	−0.22	0
14	0.77	0.9	0.16	0.16	−0.19	0
15	0.72	0.46	0.57	0.15	−0.82	0
16	0.75	0.87	0.67	0.14	−0.28	0
17	0.79	0.76	1	0.6	−0.53	1
18	0.73	0.99	1	0.4	−0.53	1
19	0.74	0.51	0.51	0.76	−0.71	1
20	0.8	0.38	0.01	0.13	−0.87	0
21	0.16	0.79	1	0.57	−0.27	1
22	0.49	0.99	1	0.5	−0.26	1
23	0.46	0.83	0.36	0.45	−0.25	0
24	0.73	0.8	0.05	0.76	−0.29	0
25	0.73	1	0.68	0.88	−0.30	1
…	…	…	…	…	…	…
203	0.2	0.13	0.07	0.41	−0.92	0

HSC—Hydrocarbon supply conditions; RC—Reservoir conditions; TC—Trap conditions; CAPC—Caprock and preservation conditions; ND—Normalized depth; 1—Productive well; 0—Non-productive well.

• AODE: Averaged one-dependence estimator method (Guo et al., 2022).

• MD: Mahalanobis distance method (Xie et al., 2011).

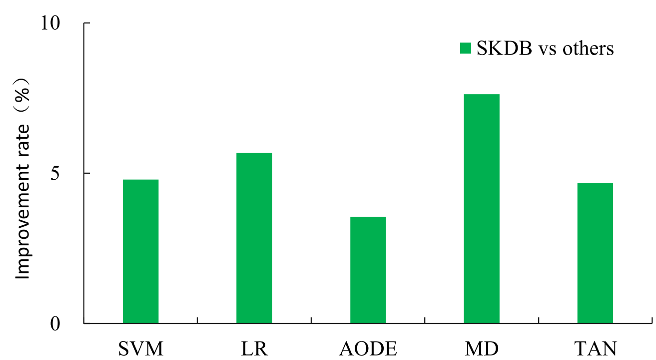

Fig. 4. SKDB's accuracy improvement results compared to other methods.

• SVM: Support vector machine method (Chen et al., 2012).
• LR: Logistic regression method (Zhu et al., 2018).
(2) Evaluation criteria

Accuracy is a common criterion to evaluate the classification performance and defined as the percentage of the correct number of samples classified to the total number of samples. Generally, the higher the accuracy, the better the evaluation method. The accuracy of a known evaluation method G for a test set $\mathbf{T} = \{\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_m\}$ with m examples, where $\mathbf{t}_i = \{x_1^i, x_2^i, \ldots, x_n^i\}$, is defined as

$$Acc(\mathbf{T} \mid G) = \sum_{i=1}^{m} \xi\left(c_i, \hat{c}_i\right) / m. \tag{13}$$

where, $\hat{c}_i$ is the class label of the ith test example predicted using method G, c_i is the true class label of the ith test example; and $\xi\left(c_i, \hat{c}_i\right)$ is the binary function. For $c_i = \hat{c}_i$, $\xi\left(c_i, \hat{c}_i\right) = 1$; otherwise, $\xi\left(c_i, \hat{c}_i\right) = 0$.

In the experimental process, for the dataset consisting of 203 exploratory wells, 90% of the samples are taken as the training set using

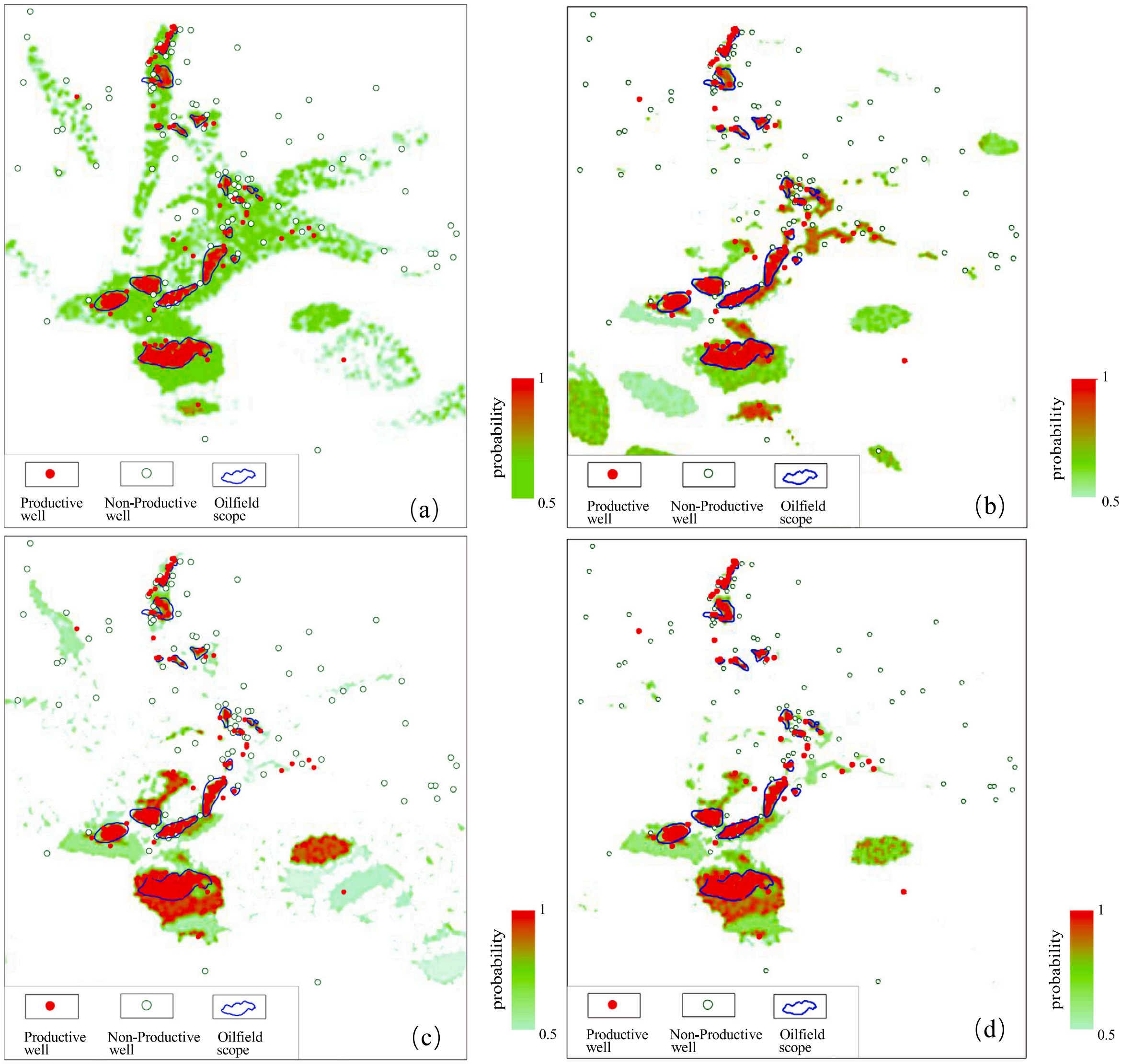

Fig. 5. Oil and gas probability prediction results ((a) MD; (b) LR; (c) TAN; (d) SKDB).

random sampling, and the remaining 10% of the samples are taken as the test set. Each algorithm was run 10 times independently, and the average of the obtained results was used as the accuracy of the corresponding method. The parameters K and the maximum number of iterations ($MaxIt$) in SKDB were set to 2 and 10, respectively. The model simplification process needs a verification set to evaluate the performance of the model, and 10% of the training samples were randomly selected as the validation set. The parameters of other algorithms are set by referring to the corresponding literatures.

3.5. Results and analysis

3.5.1. Model accuracy results and analysis

Table 3 summarizes the accuracy results of the six methods on the exploratory well dataset. As shown in table, the SKDB method proposed in this paper has the highest accuracy, reaching 87.5%, followed by the AODE and TAN methods, with the accuracy of 84.5% and 83.6%, respectively, and finally the MD, SVM and LR methods, with the accuracy respectively were 81.3%, 83.5% and 82.8%, respectively. Fig. 4 is the percentage result of the accuracy improvement of SKDB relative to other methods. As shown in figure that the accuracy of the SKDB method is improved by 4.79%, 5.68%, 3.55%, 7.63% and 4.67% compared with the SVM, LR, AODE, MD and TAN methods, respectively. Compared with other methods, there are two main reasons why the SKDB method can improve the accuracy: (i) From the algorithmic perspective, the attribute sorting process in SKDB method can capture the dependency between class variable and geological attribute variables to the maximum extent, so that the geological factors related to oil and gas can enter the model in order from strong to weak, thus effectively improving the classification performance. In the filtering process, it is ensured that the edges with strong dependencies between different geological attributes are retained in the network, while those weakly dependent edges are dynamically removed by iterative search, thus improving the classification performance of the classifier. (ii) From the data perspective: high-quality data ensures the effectiveness of the model, the key geological factors data extracted in this paper can well reflect the spatial distribution of oil and gas resources, so the SKDB method can improve accuracy. The above results and analysis demonstrate that the SKDB approach can not only effectively tackle the issue of oil and gas exploration risk prediction, but also has advantages compared with the currently popular methods.

3.5.2. Model prediction results and analysis

The SKDB model was established using the dataset of 203 exploratory wells, and the oil and gas probabilities of 12,951 evaluation units were predicted based on the model, and the oil and gas occurrence

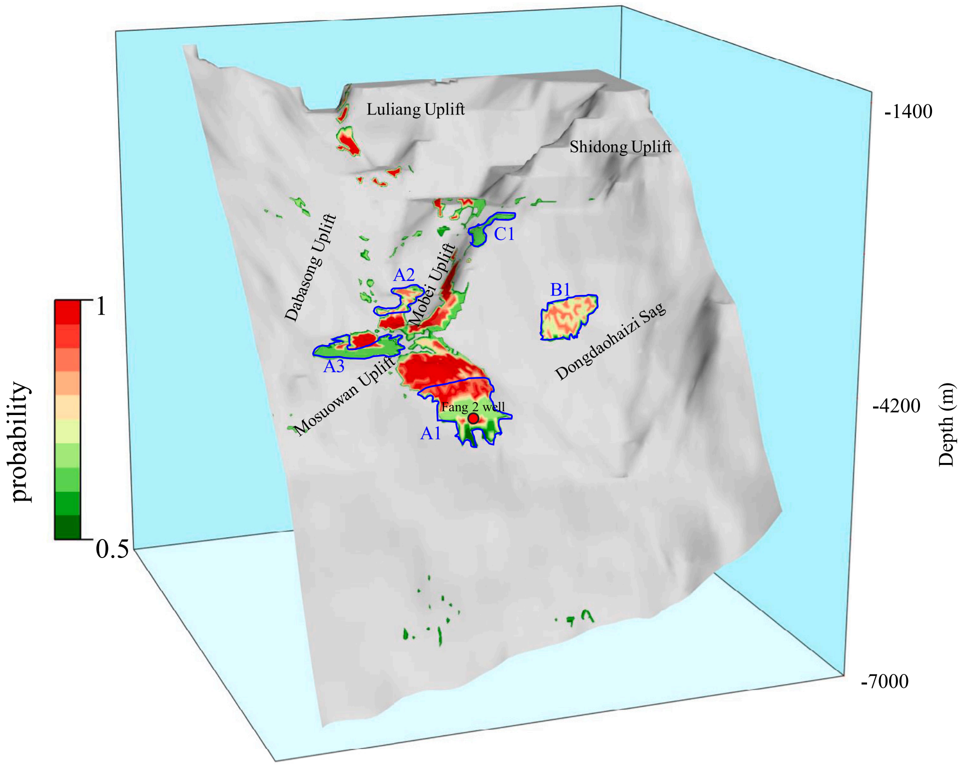

Fig. 6. Prediction of favorable distribution areas of hydrocarbon resources in the study area.

Table 2
Grid dataset.

Number	HSC	RC	TC	CAPC	ND	Well
1	0.18	0.42	0.14	0.48	−0.18	–
2	0.22	0.17	0.20	0.54	−0.40	–
3	0.79	0.10	0.04	0.89	−0.31	–
4	0.19	0.13	0.17	0.60	−0.31	–
5	0.18	0.30	0.19	0.15	−0.13	–
6	0.76	0.93	0.67	0.17	−0.22	–
7	0.17	0.94	0.44	0.72	−0.18	–
8	0.17	0.17	0.04	0.51	−0.18	–
9	0.75	0.91	0.12	0.42	−0.45	–
10	0.11	0.11	0.20	0.40	−0.40	–
11	0.18	0.42	0.02	0.59	−0.31	–
12	0.10	0.81	0.08	0.56	−0.18	–
13	0.18	0.76	0.20	0.72	−0.31	–
14	0.13	0.84	0.44	0.81	−0.18	–
15	0.12	0.47	0.03	0.17	−0.13	–
16	0.73	0.91	1.00	0.86	−0.45	–
17	0.14	0.49	0.58	0.55	−0.18	–
18	0.71	0.89	0.61	0.88	−0.27	–
19	0.48	0.39	0.70	0.58	−0.27	–
…	…	…	…	…	…	–
15951	0.17	0.80	0.13	0.53	−0.14	–

Table 3
Accuracy results of each method on exploratory well dataset.

Method	SVM	LR	AODE	MD	TAN	SKDB
Accuracy (%)	83.5	82.8	84.5	81.3	83.6	87.5

probability map was drawn. Meanwhile, three methods (LR, MD and TAN) were used as comparison methods to predict the occurrence probability of oil and gas in the Sangonghe Formation.

Figs. 5(a)–5(d) are the predicted oil and gas probability maps using the four methods of MD, LR, TAN, and SKDB, respectively. The higher the predicted probability of oil and gas occurrence, the lower the risk of petroleum exploration. As shown in the maps, the probabilities of hydrocarbon occurrence predicted by all four methods were very high over the range of discovered fields (blue curve). Meanwhile, it can be observed that the productive wells were basically situated in areas of high probabilities, while the non-productive wells were basically placed in areas of low probabilities. The above findings showed that the predicted results of the four methods were consistent with the actual exploration results in the known oil field areas. However, as shown in Figs. 5(a) and 5(b), the oil and gas distribution patterns predicted by MD and LR were obviously different from those predicted by SKDB in Fig. 5(d). The first two methods show that high hydrocarbon-bearing probability areas were widely developed in the whole area. From the analysis of distribution shape and distribution area, these areas with high oil and gas bearing probability were neither in line with the geological law nor the current geological understanding. From the perspective of oil and gas distribution patterns, Fig. 5(c) and Fig. 5(d) were similar, but there were still some differences. As shown in Fig. 3, the occurrences of oil and gas in the study area were influenced by several factors. The high hydrocarbon generation probability areas in Fig. 5(d) basically correspond to the high evaluation value areas of the reservoirs and traps. However, some high occurrence probability areas in Fig. 5(c) correspond to areas with low evaluation values of reservoirs and traps, indicating that SKDB was more sensitive to reservoir and trap conditions than TAN, and that its prediction effect was better. For example, in the lower right corner of Fig. 5(c), TAN predicts a high probability of oil and gas occurrence, but the trap evaluation values in these areas are less than 0.2, which are not conducive to oil and gas storage. Based on the above analysis, the prediction effect of SKDB was the best, and the prediction results in Fig. 5(d) were more in line with the current geological laws and actual exploration results.

3.5.3. Prediction of favorable oil and gas areas

Fig. 6 is the predicted map of favorable distribution areas of oil and gas resources based on Fig. 5(d). The gray background is the burial depth map of the Sangonghe Formation. According to the prediction results of the SKDB model, the favorable areas for hydrocarbon distribution in addition to discovered oil fields were indicated, which includes three different kinds of favorable areas

(1) Type I : This kind of favorable areas are situated surrounding the reserve area. According to the prediction results, three favorable areas were selected: (i) the A1 area on the southeast side of the Mosuowan Uplift, including the vicinity of Well Fang-2; (ii) the A2 area on the northwest side of the Mobei Uplift; (iii) the A3 area on the northwest side of the Mosuowan Uplift.

(2) Type II : This kind of favorable areas have not yet been deployed with exploratory wells, and belongs to the new frontier area that have not been drilled. According to the prediction results, one block is selected: the B1 area near Dongdaohaizi Sag.

(3) Type III : This kind of favorable areas are complex areas, although some drilling and discoveries have been made, but reserves have not been submitted. According to the prediction results, one block was selected: the C1 area on the northeast side of the Mobei Uplift.

Among the three types of favorable hydrocarbon resources distribution, the priority level of exploration direction is type I > type II > type III.

4. Conclusions

This paper proposes the use of the SKDB method to predict oil and gas exploration risk for the first time. This method improves the accuracy of the model through attribute sorting and filtering mechanism. The validity and superiority of the method are illustrated using the Sangonghe Formation in the Junggar Basin as an example, and the following conclusions are drawn:

(1) Compared with the state-of-the-art methods (SVM, LR, AODE, MD, TAN), SKDB has the highest prediction accuracy of 87.5% and an improved accuracy of 3.55~7.63%.

(2) The results of the spatial distribution of hydrocarbon resources within the reserve area predicted by SKDB were highly compatible with the exploration results. Compared with other methods, the effectiveness and superiority of SKDB method were revealed.

(3) According to the prediction results of SKDB, three types of favorable areas for oil and gas exploration are selected, namely, type I favorable area, type II favorable area and type III favorable area. The priority of the three favorable areas is type I > type II > type III. The prediction results of the SKDB method can not only reduce the exploration risk in the study area, but also provide decision-making basis for optimizing drilling strategy.

CRediT authorship contribution statement

Hongjia Ren: Methodology, Writing – original draft, Visualization, Conceptualization. **Qiulin Guo:** Acquisition of data, Funding acquisition. **Zhenglin Cao:** Acquisition of data. **Hongbo Ren:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data

Acknowledgments

Thanks to the support of the Major Science and Technology Special Project of China National Petroleum Corporation "Research on Key Technologies for Fine Exploration in Mature Exploration Areas (No. 2021DJ07)" and "Innovation Capability Improvement Plan Project of Hebei Province (22567626H)".

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